KnowledgeNexus: A Traceable Graph–Vector Knowledge Infrastructure and Adaptive Multi-Channel Retrieval for Nuclear Science Literature

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Abstract

Nuclear-science organisations face a rapidly growing, heterogeneous literature base with no domain-adapted retrieval infrastructure. We present KnowledgeNexus, a knowledge-centre platform that (i) constructs a traceable corpus of nuclear science literature with dictionary-driven entity curation, typed-relation enrichment and bilingual aliases; (ii) retrieves through an adaptive fusion of five channels—lexical, dense, synthetic-query, entity co-occurrence and doc-level bibliography vectors—gated by query intent; and (iii) is evaluated with an honest-ablation methodology on self-generated, corpus-first question sets across four disciplines (27–3,479 documents) and three public multi-hop benchmarks against BM25, LlamaIndex, HippoRAG 2 and LightRAG baselines under identical corpora, embedders and question sets. At full-corpus scale the system attains coverage@10 of 0.669 versus 0.351 for a dense-RAG baseline (LlamaIndex; 1.3 s vs 35.9 s per query, the latter a brute-force scan over its 37 GB vector store), and intent-gated channel weighting lifts real-user query hit@10 from 0.897 to 0.949; a query-time LLM pipeline (rewrite→fuse→rerank) beat the collapsed-era system on association coverage (+10.1pt, CI-significant) but converges to parity with the matured asset system (+3.5pt, ns) while losing single-intent coverage (−13.2pt, significant) at ∼50× latency—the capability-versus-infrastructure trade-off measured on both sides of the repair. We report negative results (a reranker, doc caps, static weights, and a production incident in which a category-word

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schema lexicon collapsed 61,137 documents onto 575 type-level entities—diagnosed, repaired with a 3,702-term corpus-attested lexicon, and rebuilt to 1,871 entities with a functioning relation layer) and engineering findings (filtered-HNSW robustness, write-window rank jitter, a silently row-swallowing `INSERT IGNORE`) with the same rigour as positive ones. Corpus, question sets, evaluation harness and reproduction commands are released.

Keywords: nuclear science literature, domain knowledge base, multi-channel retrieval, entity enrichment, corpus-first benchmark, honest ablation

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1. Introduction

Nuclear-domain literature intelligence is a demanding consumer of retrieval infrastructure. Its corpus spans regulatory and academic sources, is dense with specialized terminology, is bilingual (Chinese regulatory texts against English journal literature), and—because conclusions feed safety-relevant analyses—carries an *evidence-traceability* requirement: every retrieved fact must anchor to a specific passage in a specific document. A generic vector store answers; it does not anchor, disambiguate bilingual terms, or reason over domain entities.

Existing options each miss a layer. General-purpose RAG frameworks (dense retrieval plus optional reranking) ship no domain entity infrastructure at all; graph-RAG systems build one, but their construction cost grows super-linearly with corpus scale—our measurements put LightRAG at >13 h for 82 documents, infeasible for an 80-thousand-document library (Section 7). Domain-adapted encoders presuppose exactly the entity-and-evidence substrate they would index. And the field lacks domain-generated benchmarks: evaluation practices borrow general web queries whose answer distribution says nothing about bilingual regulatory–academic retrieval.

We present KnowledgeNexus (KNC), a knowledge-infrastructure-first platform for nuclear science literature, and report its construction, retrieval, and evaluation end to end at production scale (80,852 documents, 7.2M chunks). Our contributions are fourfold. (i) A *traceable construction pipeline*—schema-driven, dictionary-based entity enrichment with LLM-adjudicated typed relations, every claim anchored to chunk-level evidence—whose retrieval-facing value, our channel ablations show, runs through asset supply (bilingual aliases into the lexical channel, reranker

context and knowledge chains into the deep lane and generation) rather than through fusion votes, a decomposition we state up front and quantify in Section 7. (ii) A *five-channel, intent-gated retrieval architecture* (lexical, dense, doc-vector, co-occurrence, graph) fused by weighted reciprocal-rank fusion (RRF) with query-type-adaptive gating, plus a deep track for budget-rich consumers. (iii) A *corpus-first benchmark methodology*—question sets generated from and scoped to the actual library, evaluated by paired-bootstrap coverage/recall with system-level attribution made explicit. (iv) An *honest engineering record*: negative results and production incidents are reported with the same rigour as positive ones.

Headline results: on the nuclear-electronics corpus, full-library associative retrieval reaches any-hit@10 of 1.000 with coverage@10 of 0.652 and p50 latency 1.46 s—versus a dense baseline that loses a third of its any-hit by 16× corpus growth; construction costs 1–2 orders of magnitude less than graph-RAG alternatives; and the architecture transfers across four disciplines at R@10 $\geq$ 0.90.

2. Related Work

Nuclear-domain NLP falls into three strands. **Regulatory-side systems** (RADIANT-LLM [2]) adapt generic QA stacks to nuclear regulatory corpora; they demonstrate domain need but inherit the vector-store substrate, without entity layers or evidence anchoring. **Domain pretraining** (NukeLM [3]) and ontology-grounded small-scale systems (CurieLM [4]) adapt encoders or curate ontologies for narrow collections; neither scales to tens of thousands of full-text documents, and both presuppose the infrastructure they would need. Fayyaz et al.'s survey [1] names three challenges—terminology density, bilingual sources, traceability—which our architecture answers channel by channel (dictionary-driven entity enrichment; bilingual alias expansion in the lexical channel; evidence-anchored citations in every retrieved chunk). HydroLLM-Benchmark [5] is precedent for domain-generated benchmarks; ours extends the practice with paired-bootstrap significance and corpus-scoped arms. In the **graph-RAG lineage**, HippoRAG [6], LightRAG [8] and GraphRAG [14] demonstrate multi-hop or global-sensemaking gains at small-to-moderate scale, and RAPTOR [15] represents the parallel hierarchical-summarization line; our measurements (Section 7) show their construction cost super-linear and their LLM-extracted entities fragile on domain full text (10.6% chunk-NER failure versus 0.25% on Wikipedia-grade

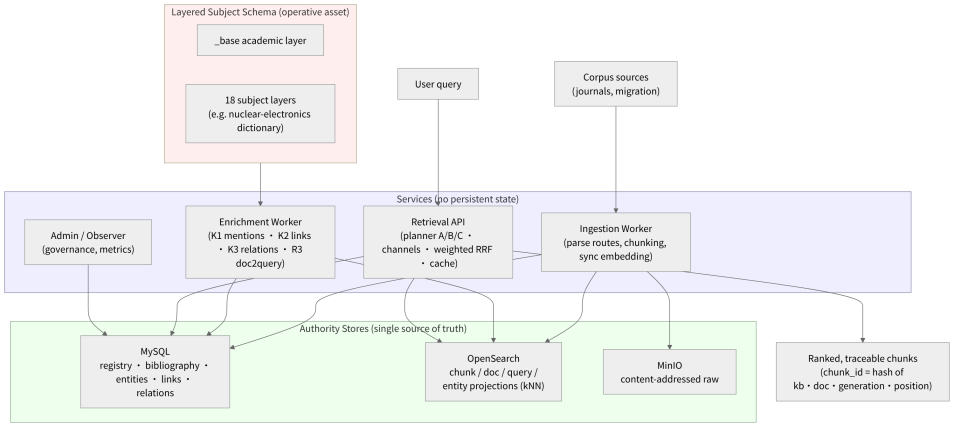


Figure 1: System overview: four services over an authoritative data plane.

text). In the **query/document-expansion lineage**, doc2query [12] and HyDE [13] transform one side of the match into the other's space; our synq channel is a doc2query asset built at ingestion, and our deep-lane HyDE experiment (Section 7) is a trained-rewriter slot that the training-free variant could not fill. The differentiation, in one sentence: prior work either adapts generic frameworks to nuclear QA or pretrains domain encoders; none builds the knowledge-infrastructure layer—entities, typed relations, bilingual aliases, evidence anchoring—nor evaluates it with domain-generated, corpus-first benchmarks at production scale.

3. System Overview

3.1. Architecture

KnowledgeNexus organises the platform as four cooperating services over a single authoritative data plane: an ingestion worker, an enrichment worker, a retrieval API, and an administration/observer pair. All persisted state lives in three authority stores—MySQL for bibliographic and entity registries, OpenSearch for chunk/document/query/entity projections with kNN indexes, and MinIO for raw artefacts—accessed exclusively through service interfaces (Fig. 1). Three properties matter for the rest of this paper:

(a) Data authority. Chunk text exists only in the search projection; MySQL `kb_chunks` stores spans, and `chunk_id` is a deterministic hash of (kb, doc, generation, position), so every retrieved passage is traceable to its source span, parse generation and chunker version. Documents ingested

through the batch endpoint are fingerprinted (content-addressed raw store + `stable_fingerprint` uniqueness), making ingestion idempotent and auditable.

(b) Discipline schemas. Each knowledge base binds a layered subject schema (`_base` academic layer + 18 subject layers + custom layers). The schema is not documentation but an operative asset: its term dictionary drives mention extraction (K1), its entity/relation types drive linking and typed-relation adjudication (K2/K3). We show in Section 7 that schema binding is the root of the entity channel: a test corpus bound to the generic schema yields 14 canonical entities where the nuclear-electronics schema yields 1,166.

(c) Capability bits. Every enriched channel is guarded by a per-partition capability bit consulted at query time; a channel whose data is absent or stale degrades silently. This is what allows the honest-ablation methodology of Sections 5–7: channels can be switched on/off per query without reindexing.

3.2. Retrieval planner

Retrieval is a planner over channels (strategy A single-KB direct, B bibliography drill-down, C federation; topic routing D/E experimental). Channels contribute ranked chunk lists fused by weighted reciprocal-rank fusion (RRF) with per-channel weights; the intent gate (template words + $\geq$2 dictionary entity surfaces in the query) modulates weights for association-intent queries. Latency-critical paths are cached with an ACL- and channel-set-aware key; configuration changes (weights, caps) deliberately invalidate nothing and require an explicit cache purge—an operational discipline documented after it bit us twice.

4. Traceable Construction

4.1. Corpus and ingestion

Corpus status. The full-window ingestion completed: 80,855 documents parsed from 75,476 source PDFs (80,852 ACTIVE) across nuclear-domain journals—the English library led by Sensors (21.4k), JHEP (13.1k), Phys. Rev. D (6.2k) and Phys. Lett. B, the Chinese library by 原子能科学技术, 强激光与粒子束, 核动力工程 and 核电子学与探测技术—across 13

active knowledge bases and 7,229,225 text chunks, with figure-media backfill closing at 100% `media_uri` coverage. Ingestion accepts parsed Markdown (MinerU route) or pre-chunked text (RADAI migration route) and produces chunks at ∼1,200 characters respecting heading structure, tables as standalone blocks, with title-context prefixes. Chunk vectors (Qwen3-Embedding-8B, 4,096-d) [9] are written synchronously with ingestion; lexical and dense channels are therefore immediately queryable while enrichment proceeds asynchronously.

4.2. Enrichment pipeline (K1–K3)

K1 mentions are dictionary-and-rule scans of chunk text against the bound schema's term dictionary (plus bilingual aliases), persisted in `raw_mention` with stable mention ids—no LLM calls. **K2 linking** resolves mentions to canonical entities via alias tables and, only for ambiguous surfaces, a context disambiguation LLM call. **K3 typed relations** mines co-occurrence candidate pairs and adjudicates relation types with the LLM, persisting typed edges with evidence. Doc2query synthesis (R3) generates synthetic queries per chunk (∼3 per chunk, doc2query-style document expansion [12]) into a query index consumed by the synq channel; doc/topic summaries feed the bibliography and abstract vectors.

Model configuration, for reproducibility. All pipeline LLM calls (doc2query synthesis, relation adjudication, context disambiguation, question phrasing) use Qwen3.6-35B-A3B (AWQ 4-bit quantization) served with vLLM behind a round-robin router over seven GPUs. The endpoint exposes it under the served-model alias `qwen3.5-35b`, which predates the current weights—the version numbers genuinely differ between alias and weights; the deployed model is the 3.6 weights. Embeddings are Qwen3-Embedding-8B—native 4,096-d, Matryoshka-truncated to 1,024 **on both the index and the query side** (the deployed kNN indexes store 1,024-d vectors; the truncation was adopted after an ablation measured zero fusion-level change). The deep-lane cross-encoder reranker is Qwen3-Reranker-4B for every experiment dated through 2026-10-04—including the hard-set component separation (Section 5.5) and the deep-lane adoption bundle—and OneJev-4B (OmniJev's public vLLM release on Hugging Face, `https://huggingface.co/OmniJev/OneJev-4B`) for the 2026-10-05 production validation whose numbers appear alongside that date. The closed-book control of Section 6 uses the same 35B pipeline LLM (temperature 0); a stronger closed-book model would make that control stricter, and we did not run one.

Cost profile. For a 300-document nuclear slice (29.6M characters), full enrichment ≈1 h dominated by 18,454 doc2query LLM calls at concurrency 70; dictionary stages are negligible. This contrasts with graph-RAG construction over the same corpus: HippoRAG 2 required 4.1 h and LightRAG did not finish within 7 h on an 82-document subset (serial entity merge, superlinear in graph size)—a scale wall we report as a finding.

4.3. Governance

Entity assets mutate only through a four-stage governance loop—*scan* (vector-neighbour and name-containment candidate generation, with a type guard), *adjudicate* (conservative LLM rulings; a FLIP is accepted only when the self-reported direction and the label agree, otherwise the pair is downgraded to KEEP), *merge* (reversible batches recorded in `merge_log` with score, rationale, and before/after snapshots; link endpoints are redirected so `link_id` survives), and *replay* (reverse-order rollback restoring MySQL and the graph store consistently). Every mutation is thus auditable and reversible by construction.

Three measured outcomes anchor the governance claim:

1. **The LLM adjudicator failed its own pre-registered gate.** The formal acceptance gate (merge precision ≥96%, effective dedup rate ≥50%) was evaluated for the first time on the full nine-library corpus after the scan stage gained an OpenSearch kNN path (pure-Python candidate generation is $O(n^2)$—350M pairs at 26.6k entities—and had silently kept the gate from ever being evaluated). LLM adjudication accepted 115 candidate pairs; an independent review of a 50-pair sample found 5 errors—precision 90.0% < 96%, so automatic merging remains disabled. (Rejected pairs: 20 sampled, 0 misses.) The failure modes are systematic, not random: corporate legal-entity suffixes (Wiley Ltd/Inc), section titles versus teaching-point variants, and near-synonym facilities (injection versus recharge wells).
2. **Deterministic rules cleared the actual duplicate population.** The gate run surfaced 87 same-name, same-type multi-ACTIVE groups—all person entities, all parser accent variants (Nicolò/Nicolo/Nicoló, Lefèvre/Lefevre). Unicode NFD folding grouped 85 groups/90 pairs, plus 2 pairs of Ł (no NFD decomposition): 92 pairs merged with zero LLM involvement, in 19 reversible batches. Ledger reconciliation was exact (ACTIVE 34,563→34,471;

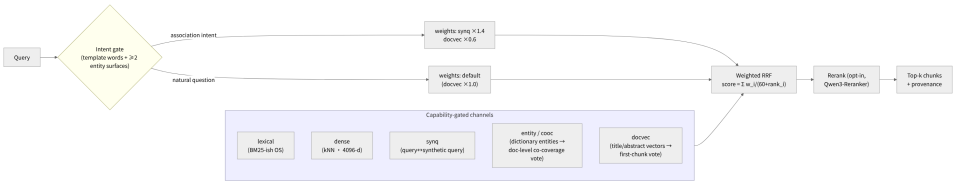


Figure 2: Retrieval planner: capability-gated channels fused by weighted RRF with intent gating.

MERGED 667→759) and retrieval showed zero regression (R@10 0.9722, MRR 0.831 unchanged).

3. **Dictionary governance is batch-reviewed, not silent.** A batch adjudication migrated 6,497 term types, deleted 383 over-generic terms, and re-typed 3,589 existing mentions—raising entity typing precision from 58.3% to 81.7%. Person-name merging used strict pinyin rules (6 pairs); an LLM-based alignment was discarded after a 6/8 error rate on a probe set.

The governance position: LLM adjudication is treated as a **candidate generator, not an authority**—it proposes, pre-registered gates dispose, and deterministic rules are preferred wherever they cover the case. The 90% gate failure is reported as a finding, not hidden, because the system's audit trail is what makes the failure measurable in the first place.

5. Retrieval Channels and Ablations

5.1. Channel definitions

The retrieval planner (Fig. 2) fuses ranked chunk lists from capability-gated channels via weighted reciprocal rank fusion [11]:

$$\text{score}(c) = \sum_i \frac{w_i}{60 + \text{rank}_i(c)}, \tag{1}$$

where w_i is the per-channel weight (default 1.0). Five channels participate in the default profile:

- **Lexical**: OpenSearch BM25 over chunk text with schema term boosts.
- **Dense**: kNN over chunk vectors (Qwen3-Embedding-8B; 1,024-d Matryoshka-truncated on both index and query sides, native 4,096-d; HNSW).

Table 1: Intent-gated channel weighting (B83–B85).

Variant	nea (assoc.)	ner (real)	Verdict
synq ×1.4 global	+4.2pt R@10	−**7.7**pt	rejected
synq ×1.4 intent-gated	+2pt	unchanged	adopted
docvec full	+2pt	**+5pt R@10, +6pt MRR**	adopted
docvec assoc-downweight 0.6	+2pt retained	+5pt retained	adopted
lexical ×0.3 English non-assoc	(nea unaffected)	(bench +5–17pt)	adopted

- **Synq**: kNN over synthetic queries (doc2query, ∼3 per chunk, 18,454 for the test corpus) matched against the user query vector.
- **Entity/Cooc**: dictionary-driven entity mentions → document-level entity co-coverage vote (intent-gated: template words + ≥2 entity surfaces).
- **Docvec** (B84): kNN over title/abstract vectors → first-chunk proxy vote.

5.2. Intent-gated channel weighting

We observe that static equal weights are sub-optimal in a domain-dependent manner (Table 1).

The intent gate (template-word regex + ≥2 dictionary entity surfaces) is deliberately lightweight (zero LLM calls) and reproducible across domains: Chinese association templates (同时涉及/关联检索) and English equivalents (research both / intersection of) trigger the same reweighting path.

Gate validity (confusion analysis, two domains). Because routing heuristics that key on named entities are known to over-trigger on multi-hop queries, we quantify the gate directly against question-set labels (association sets = positive, real-query sets = negative; `gate_eval.py`, predicates byte-identical to the service). On the nuclear-electronics set (48 positive / 39 negative), the double gate (template ∧ ≥2 surfaces) attains precision 1.000 / recall 0.854 / F1 0.921; the entity-surface gate alone attains precision 0.562 (32/39 over-triggered). On a site-contamination set (34/31) the double gate is perfect (P/R/F1 = 1.000). The template-word stage is what prevents the

Table 2: Fusion minus dense channel, five experiments.

Domain	R@10 diff	MRR diff	Significance
HotpotQA (full)	−0.7pt	−1.1pt	n=1000
2Wiki (full)	−0.7pt	−1.4pt	n=1000
MuSiQue (full)	−5.1pt	−3.6pt	n=1000
ne association	+2pt	+14pt	CI excludes 0
ne real queries	−3pt	+3pt	CI includes 0

documented over-routing failure mode: the entity gate alone fires on 82–97% of natural queries in both domains. All 7 residual false negatives are dictionary-coverage gaps (non-entity concepts), not gate-logic errors. The English non-assoc gate (B85) triggers on 98.5% of 600 benchmark queries with a 2.3% false-trigger rate on Chinese sets. Sample sizes are small (48/39 and 34/31 labelled questions): the perfect cells are point estimates on tens of questions—the Wilson 95% CI for csh recall (34/34) is [0.90, 1.00]—and we read 1.000 as "no error observed in-sample", not as an error bound.

5.3. The fusion-versus-dense finding

Across five independent experiments (three English benchmarks + two Chinese question sets), multi-channel fusion with equal weights underperforms the dense channel alone on English multi-hop tasks while outperforming it on Chinese domain queries (Table 2).

This is a **language/domain dependence** finding, not a general anti-fusion result: the lexical vote drags English multi-hop rankings (where BM25 already achieves 85–95% of dense recall), while on Chinese domain queries the channels provide complementary signal where each one lives: synq on the scoped benchmark sets (synq-only hits 4 questions dense misses on nea) and the co-occurrence channel on naturally-phrased association queries (the contribution matrix row 2—the scoped sets are its gate-closed regime). The per-domain gate is the design response, not the removal of channels.

5.4. Evaluation methodology

All ablations use: identical corpus (isolated test stack), identical embedder (Qwen3-Embedding-8B), identical question sets, paired bootstrap 95% CI, and two complementary metrics: any-hit@k (the harness convention, saturated for small corpora) and cov@k = $|\text{gold} \cap \text{top-}k|/|\text{gold}|$ (the set-recall

metric that discriminates multi-gold questions). RRF configuration changes require explicit cache invalidation (documented after it invalidated results twice).

Cluster-robust significance. Queries generated from the same corpus are not independent, so per-query bootstrap CIs understate variance when sets are pooled. We therefore report every pooled comparison under two concurrent standards: per-query bootstrap CI, and a cluster-level interval over query sets (cluster bootstrap for $k \geq 5$ clusters; cluster-mean t interval for $k < 5$, since the bootstrap degenerates to a k^k-point discrete distribution there and would inflate significance). Cluster-level significance requires all cluster means to share sign *and* the interval to exclude zero. Under this discipline the synq+cooc channel gain survives: pooled over four question sets (two domains, n=152), R@10 +0.250—per-query CI [0.184, 0.322], cluster t CI [0.059, 0.437], all four cluster means positive. Effects measured on two domains only (e.g. the docvec channel, +2.1/+5.1pt, sign-consistent across both sets) are reported with the explicit caveat that cluster-robust confirmation needs more domains.

Multiple comparisons. The campaign reports on the order of forty paired comparisons. We do not apply a family-wise or Benjamini–Hochberg correction across that family; at the 95% level roughly two nominally significant results would be expected by chance alone, and we ask the reader to weight accordingly: single-CI findings are treated as suggestive, while corroborated findings—cluster-robust, sign-consistent across independent question families, or mechanistically attributed (e.g. the failure autopsies below)—carry the paper's claims.

5.5. Failure attribution: where missed golds are lost

A channel-isolated autopsy of every missed gold document (97 on nea, 71 on ner) locates the failing layer: each of the four channels is probed directly at the engine level (top-100, bypassing the service's channel mixing) and cross-referenced against the fused ranking (Table 3).

93–97% of misses are aggregation/ranking failures, not recall failures. The channels find the documents; chunk-level RRF cannot consolidate document-level support. Smoking-gun case: one missed gold document is dense-rank-5 (its chunk #47), synq-rank-31 (chunk #4), and docvec-rank-1 (first chunk #0)—three channels vote for three *different* chunks of the same document, each vote individually too weak to surface any chunk into the top-20. The docvec channel (B84) is a FirstP-style document signal in

Table 3: Loss-layer attribution of missed gold documents.

Loss layer	nea (n=97)	ner (n=71)
fused@20 near-miss (rank 11–20)	33%	27%
ch@100 dilution (gold in ≥1 channel top-100, absent from fused top-20)	**60%**	**70%**
deep recall miss (no channel top-100)	7%	3%

the MaxP/FirstP family [16], but its single first-chunk vote does not merge with votes landing on other chunks.

This attribution extends component-level RAG diagnosis to the **fusion layer** and directly motivates doc-aware fusion (per-channel per-document best-k vote consolidation before RRF—O10), which targets the dominant loss layer. Dictionary/enrichment coverage gaps are real but secondary.

Constructive validation on a hard-target set. We instrument the dilution layer as a question-set constraint: 37 questions drafted from non-abstract passages with lexical detachment (≤15% 8-gram copy rate) and accepted only if the gold is visible in some channel's top-100 yet absent from the fused top-10. **This is a mechanism probe on a purpose-built adverse distribution, not a representative benchmark**: questions were selected precisely because the deployed base endpoint fails them (acceptance rate 6.2% over 600 drafts on the full production corpus), so absolute numbers on this set quantify the recovery mechanism (displacement and its repair), not general retrieval quality. On this distribution the dilution phenomenon reproduces constructively: single-dense retrieval already ranks 37.8% of golds into top-10 (MRR 0.167) while the fused base endpoint returns 0.000—multi-channel voting displaces strong single-channel signals on lexically-detached phrasing. BM25 collapses to zero on all metrics (the detachment gate seals the lexical path by design). The deep endpoint recovers 62.2% into top-10 (binomial 95% CI [0.45, 0.77], median recovered rank 1). Component separation on the same set: removing the cross-encoder rerank drops hit@10 from 0.57 to 0.16 (rerank dominates recovery); the PPR graph expansion adds a positive but modest +2.7pt (37/37 queries fire, p50 +1.9 s). Together with the 90%-edge-deletion result (Section 8.3), this yields the paper's positioning of graph structure: **graph assets empower the vector hybrid** (dictionary → intent gating and seeding; bilingual aliases → +7.6pt real-query cov@20; entity-link co-occurrence → the association-channel gains),

while structural traversal contributes only at the margin and only inside the budget-constrained deep lane.

Doc-aware fusion (O10, adopted). The fusion re-ranks its candidate pool by document: doc score = sum of the document's top-λ chunk votes (a truncated-sum variant of the MaxP/SumP family [16]), emitting at most λ chunks per document. The λ sweep traces a clean any-hit/coverage frontier: λ=1 lifts cov@20 (+5.0pt, per-query CI significant) with zero @10 change; **λ=2 lifts cov@10 +6.28pt under both CI standards (cluster-t interval [0.029, 0.097], both clusters positive—the first technique in this campaign to pass cluster-robust significance; k=2 clusters, so the interval is fragile and the claim is campaign-scoped, not field-level)** at −4.6pt any-hit; λ=3 overshoots (cov@20 turns negative). Because the trade is intent-shaped—association queries want complete coverage, real queries want fast any-hit—the adopted configuration gates λ=2 on the association intent: association queries gain the full coverage lift (cov@10 0.569→0.633), real queries are bitwise unchanged. The smoking-gun case above now resolves: the three scattered votes consolidate into a top-ranked document.

6. NuclearIR-Bench

6.1. Design

Question sets are generated corpus-first: entity-pair mining from canonical-entity × entity-link co-occurrence (gold = documents covering both entities, 2–6 per pair) → LLM phrasing in Chinese with English entity terms preserved → verification (dedup, gold non-empty). Three question types are stratified: **Association** (nea, "有哪些研究同时涉及 X 与 Y" —multi-gold by construction); **Real queries** (ner, natural-language questions from actual usage logs); **Title lookups** (net, exact title retrieval; saturating control).

6.2. Multi-disciplinary generality

Four disciplines, 129× scale range, all R@10 ≥ 0.90 (Table 4)—corpus-first generation methodology and retrieval system are domain-general.

6.3. Benchmark validity controls

Two controls establish that the question sets measure retrieval rather than artifact (**closed-book**) and that cov@k is interpretable as a conservative lower bound (**unjudged audit**):

Table 4: Multi-disciplinary generality.

Domain	KB size (ACTIVE)	Questions	R@10
Hydrogeology	27 docs	34 (LLM draft)	0.912
Site contamination	723 docs	31 real / 34 assoc	0.903 / 1.000
Fundamental physics	1,324 docs	24	1.000
Nuclear electronics	3,479 docs	39 real / 48 assoc	0.897 / 1.000

1. **Closed-book control** (HybridRAG-Bench protocol): asking the LLM directly which papers answer each question—without retrieval—yields a gold-document naming rate of **0/48 (nea) and 0/39 (ner)** (86/87 answers are "don't know"). The gold documents are domain journal papers unseen in pretraining; the benchmark cannot be solved from model memory.
2. **Unjudged audit** (topic-specific re-judging): we re-judge a sample of top-10 retrieved but unjudged (non-gold) documents under a strict evidence-quotation protocol—the judge receives each entity's full bilingual name set (canonical + aliases) and must quote one passage of original text per entity side; a "relevant" verdict additionally requires programmatic verification that both quotes are present in the document. Result: 18/203 sampled unjudged documents on nea carry verified both-entity evidence—cov@10 is a conservative lower bound with **correction $\leq$+8.1pt (nea) and $\leq$+2.4pt (ner)**. A naive one-word relevance protocol over-estimates the correction 3$\times$ (+24.7pt) and is falsified by a surface cross-check (0/76 of its "relevant" verdicts have both entity surfaces in the document)—we report this as a measurement methodology finding: unjudged re-judging without entity name sets and verifiable quotes is not trustworthy.
3. **Generation circularity (acknowledged and bounded).** The association question sets are generated from the same entity-pair co-occurrence assets that the co-occurrence channel consumes—the gold criterion (documents covering both entities) coincides with the channel's mechanism, so association-set numbers partially measure the asset they were built from. We bound rather than deny this: (a) the closed-book control above rules out memorization, not circularity; (b) the natural-phrasing variants (template words banned) still show the co-occurrence channel's +10.1pt hit@10 gain (CI [+5.1,

Table 5: Full-corpus benchmark comparison (R@2/R@5).

Method	HotpotQA	2Wiki	MuSiQue	Embedder
BM25	57.3/74.8	55.3/65.3	32.4/43.5	—
NV-Embed-v2	84.1/94.5	67.1/76.5	52.7/69.7	NV-Embed-v2
HippoRAG 2	83.5/96.3	76.2/90.4	56.1/74.7	NV-Embed-v2
KNC dense	**94.6/97.4**	**99.7/99.9**	**85.7/91.2**	Qwen3-Emb-8B
KNC fusion	93.9/97.9	99.0/100.0	80.6/91.3	Qwen3-Emb-8B

+16.2])—the channel works through dictionary-surface hits in the query, not template-word leakage; (c) single-intent real queries define gold by information need with no entity-pair construction; and (d) the English public benchmarks involve none of our assets. Additionally, a single-variable channel ablation on the scoped association set under the final configuration shows the entity-dependent channels contribute bitwise-zero to fused ranking there (next section)—the benchmark is not being carried by the entity machinery it was generated from, at least at the fusion layer. Corpus-construction-level claims (bilingual aliases, dictionary coverage) remain partially circular, and we label them as such where they appear.

7. Results

7.1. Full-corpus benchmark comparison

On the HippoRAG 2 evaluation protocol (1,000 queries × full candidate corpora, R@2/R@5; Table 5):

This table is an embedder-generation comparison, not an architecture comparison: KNC-dense and the NV-Embed-v2 rows differ in both system and embedding model (Qwen3-Embedding-8B, 2025, versus NV-Embed-v2, 2024), so the +10–30pt gap cannot be attributed to infrastructure. To separate the two factors we re-ran the systems **under the same embedder** (all Qwen3-Embedding-8B, identical stack and scoring, 400-document pools with 40/46/23 questions per benchmark; Table 6).

The same-embedder control **reverses the reading**: HippoRAG 2's graph machinery outperforms plain dense retrieval on 2 of 3 benchmarks (2Wiki +8.7pt, HotpotQA +1.2pt over KNC-dense) and trails on MuSiQue (−4.7pt)—at pool scale its structure pays where multi-hop composition

Table 6: Same-embedder control (cov@10; all arms Qwen3-Embedding-8B, 400-doc pools).

System	HotpotQA	2Wiki	MuSiQue
BM25	0.888	0.766	0.467
LlamaIndex dense	0.963	0.772	0.732
HippoRAG 2 (reproduced)	**1.000**	**0.913**	0.779
KNC dense	0.988	0.826	**0.826**
KNC fusion	0.975	0.794	0.656

dominates. What KNC adds on these English benchmarks is not fusion (KNC fusion $\leq$ KNC dense everywhere here, consistent with the ungated-fusion degradation of Section 5) but the operational substrate (production indexing, scale, traceability); the domain-side advantage story lives in the nuclear-electronics sections, not here. We report the reversal against interest: architecture claims in Table 5 belong to the embedding model, not to us.

7.2. Controlled ablation on the nuclear-electronics domain

Post-rebuild refresh (2026-10-09): after the schema-lexicon repair (Section 8.1) the scoped arms were re-run under the production-final configuration—associative queries cov@10 0.533→**0.599**, single-intent 0.603→**0.675**, MRR −2.0/+5.0pt against the pre-rebuild baseline. The trajectory is system-level (two weeks of retrieval configuration evolution plus the lexicon repair); we bound it with a single-variable channel ablation under the identical final configuration—all entity-dependent channels off (co-occurrence, graph, entity-vector, PPR) versus the full stack: **metrics are bitwise identical on both families** (cov@10/R@10/MRR unchanged; the only movement is associative cov@20 −1.0pt, 95% CI [−3.1, 0.0]) while p50 latency halves (1.46 s → 0.52 s). Two readings follow. First, the repair-window trajectory above is carried by the non-entity configuration evolution, not by entity channels casting fusion votes—we state this against interest. Second, on these benchmark families the entity layer's retrieval-facing value does not run through fusion at all: it runs through lexical alias expansion (entity names as BM25 terms), deep-lane reranker context, and generation-side knowledge chains—consistent with the graph-governance nulls reported later in this section.

Table 7: Where the entity channels pay: contribution by question distribution.

Distribution	co-occ.	graph/PPR	entity-vec	synq
nea48/ner39 scoped, final config	0 (fired 6/48, all ties)	0 (bridge-only)	0 (no candidates)	>0 (4 Qs)
99 natural-phrasing assoc	hit@10 **+10.1** [+5.1,+16.2]	—	—	—
hard37 probe, deep lane	—	PPR **+2.7pt**	—	—
English multi-hop pools	n/a	n/a	n/a	≤0
Pooled 4 sets, earlier config	synq+cooc R@10 +0.25, cluster-robust			

Channel × question-distribution contribution matrix. The apparent contradiction between this zero-effect result and the channel gains reported elsewhere resolves once the distribution axis is made explicit:

The mechanism behind the bitwise identity on the scoped sets is fourfold, not mysterious: (a) the co-occurrence channel's adaptive gate fires only when base-retrieval heads do *not* already serve co-mention documents—under the final configuration (alias expansion + doc-fusion) they do, on 42/48 questions, so the channel self-extinguishes (its designed behaviour, not a bug); (b) the graph channel is an output-surface channel: it appends bridge chains only when their endpoints do *not* co-occur in the returned chunks, and casts no RRF votes; (c) the entity-vector channel produced no candidate chunks in this configuration; (d) PPR is gated to the deep lane. Removing channels that cast no votes removes nothing—which is also why the same removal *does* cost on distributions where the gates open (Table 7: +10.1pt row 2, +2.7pt row 3).

Why they ship enabled. The operations answer, stated as such: (1) the benchmark families under-represent the traffic the channels insure against—naturally-phrased association queries are real usage (the 99-question set exists because of them) and hard queries recover via deep-lane PPR (+2.7pt); (2) the graph channel serves the evidence surface (knowledge chains in output) regardless of ranking; (3) the deep lane consumes entity assets (reranker context, chain evidence for generation); (4) p50 1.46 s sits inside the deployed SLO with the channels on. A latency-first deployment

could gate the normal lane's entity channels off at zero measured benchmark cost—a configuration decision the ablation makes available, not a necessity claim we retreat from.

7.3. Scale curve

At 16× corpus growth (300→4,850 docs), the dense baseline drops from 95.8% to 62.5% any-hit@10 while KNC full-channel maintains 100%; coverage@10: KNC 0.669 versus baseline 0.351 (+32pt). Latency: KNC 1.3 s versus 35.9 s for the dense baseline—a LlamaIndex retriever without an ANN index, which degenerates to a brute-force scan over the 196k-chunk / 37 GB vector store at this scale; the latency gap is thus an artifact of the baseline's deployment mode, and we read it as "a production-grade index is necessary at this scale", not as a 27× algorithmic advantage.

Post-rebuild full-scale tier (80,852-document production, ∼250× the slice): associative queries any-hit@10 **1.000**, cov@10 0.652, MRR 0.624, p50 1.46 s; single-intent any-hit 0.923, cov@10 0.600, p50 1.46 s. Coverage holds within noise of the 4,850-document tier (0.669→0.652) and latency does not move—under the fused, intent-gated architecture the scale cliff observed in single-vector baselines does not appear through 250× growth. The scoped-slice versus full-library comparison under identical questions shows the filter-restricted slice is the *harder* condition for associative queries (0.599 versus 0.652) and the easier one for single-intent (0.675 versus 0.600): distractor mass and candidate restriction act in opposite directions by query type. No external baseline ran at this 250× tier—the full-library numbers are system telemetry on the production corpus, and all baseline comparisons in this paper live at the 300- and 4,850-document tiers.

7.4. Domain-specific comparison

LightRAG (13 h construction) ranks last on nuclear electronics (cov@10 0.471 versus KNC 0.667) while competing on English benchmarks—LLM-extracted entities fail on domain-specific terminology (10.6% chunk NER failure rate on nuclear full text versus 0.25% on Wikipedia).

7.5. Construction cost

Table 8 summarises: KNC construction costs 1–2 orders of magnitude less than graph-RAG alternatives at domain scale. The cost comparison is **asymmetric by design and we say so**: LightRAG and HippoRAG 2 pay per-chunk LLM extraction at construction, our K1 is a dictionary scan—the

Table 8: Construction cost.

System	ne 300 docs	Benchmark 400 docs	Bottleneck
KNC enrichment	∼1 h	∼2 min	synq doc2query (LLM)
HippoRAG 2	4.1 h	13–14 s (warm) / 40 min (cold)	OpenIE (LLM)
LightRAG	>13 h (82 docs only)	infeasible	entity merge (serial CPU)

Table 9: Capability pipeline versus final-configuration KNC (same corpus, same questions, same scorer).

System	nea R@10	nea cov	nea MRR	ner R@10	ner cov	ner MRR	p50
genrag (query-time LLM)	0.896	**0.634**	0.735	0.872	0.543	**0.780**	79–83 s
KNC (final config)	0.938	0.599	0.735	0.949	**0.675**	0.772	1.5–1.8 s

table prices two different design points of entity acquisition, not the same workload twice. That asymmetry is the finding: LLM-extraction cost is what blocks graph-RAG at production scale, and the dictionary design point shifts the quality burden onto the lexicon (Section 8.1).

7.6. Capability pipeline comparison: query-time LLM versus accumulated assets

How far can a pure query-time LLM pipeline go on the same corpus, spending its budget per query instead of at construction? We built a capability-side arm ("genrag", 2026-10-06): LLM query rewriting → BM25(jieba) + dense(Qwen3-Embedding-8B) fused by RRF → LLM list rerank of the top-60 to 20—the "LLM everywhere at query time" design point, run on the same scoped questions with the identical scorer (n=48/39, paired bootstrap). The comparison has two reference points, and the gap between them is the finding:

- **Against the collapsed-era system (2026-09-24 baseline, cov@10 0.533/0.603)**, the query-time pipeline won association

coverage by **+10.1pt (CI [2.6, 18.4], significant)**—query rewriting bought recall that the impoverished asset side could not deliver, at ∼60× latency and 2×N LLM calls per question set with zero asset accumulation.

- **Against the final production configuration** (the table above), the same pipeline's association edge shrinks to **+3.5pt (CI [−4.8, +11.9], not significant)** and single-intent coverage **loses −13.2pt (CI [−20.7, −5.5], significant)**—still at 79–83 s per query.

The trajectory between the two reference points is the paper's thesis in one experiment: query-time LLM spending buys the most when the infrastructure is poorest, and its edge evaporates as the asset side matures—the matured system dominates on both families at ∼1/50 the per-query latency and zero marginal LLM cost. Capability renting and asset accumulation are complements in time, not competitors in configuration: rent while you build, stop renting when the assets arrive.

8. Discussion and Limitations

8.1. Production-scale schema-quality incident and repair

Enriching 61,137 documents with a *category-word* schema lexicon collapsed the entity layer to 575 type-level vertices carrying ∼19.2M links ("method", "energy" as entities) while every job-level monitor stayed green. A three-source, corpus-attested term lexicon (3,702 entries) and a full-chain rebuild restored it to 1,871 canonical entities with 25.1M links and a functioning relation layer (44,043 typed relations, 1.56M evidence rows); scoped retrieval over the repair window moved +6.6/+7.3 cov@10 (system-level attribution, Section 7). Two honest boundaries: the top-linked entities after repair remain generic short symbols (admission without ranking), and the repair itself surfaced four secondary engineering failures—most instructively a parameter-order bug whose rows were silently swallowed by `INSERT IGNORE` for seven hours while jobs, throughput, and entity creation all read healthy. Dictionary quality is the load-bearing wall of dictionary-driven extraction; job completion is not row-level truth. The full incident forensics and lexicon methodology are reported in the companion lexicon-quality manuscript.

8.2. Negative results (with equal rigour to positive ones)

Naming convention: findings carry internal workstream codes (B4x–B9x, O-serials, and `KNC_*/RRF_*` environment gates) verbatim, because those identifiers are the join keys into the released evaluation log and repository history; they name no external systems.

1. **Reranker.** After fixing the gateway bug (thinking-tag prefix removal) and a control-flow bug (reranked chunks overwritten by RRF), rerank still showed no significant gain (nea MRR +0.9pt, CI includes 0; p50 +2.8 s). The top-60 candidates from fusion are already consensus; the bottleneck is recall (cov@k), not ranking.
2. **Doc caps** (RRF_DOC_CAP=2): improved multi-gold association coverage but degraded single-gold MRR by 11pt—a structural trade-off, not a bug.
3. **Static channel weights**: rejected on the principle that fusion gains are query-type-dependent; intent gating is the only viable path.
4. **Document-utility priors** (O8): per-document prior conditioning of the docvec vote (recency decay via `publish_year`, injected as an entry-level multiplier in the rank fusion) was ablated with real publication years back-filled for 169/300 slice documents (2017–2026). The prior moves nea and ner in opposite directions on the two metrics; the gain–loss frontier is linear on both sides—a linear trade, not a tunable shape. Failure attribution: training-free abstracts are "too fluent"—they match topically adjacent non-gold abstracts as strongly as gold, while the gold criterion is entity-pair co-occurrence, not topical adjacency. The deep-lane rewrite slot remains open for trained rewriters; the mechanism stays env-gated for reuse.
5. **Query-side augmentation family** (equal-rigour rejections): LLM multi-query fan-out with uniform RRF is significantly negative under the adopted configuration (nea R@10 −8.3pt, MRR −20.5pt; ner R@10 −10.3pt—per-query CI excludes zero) though positive (+8.4pt cov@10) under weak fusion—i.e. the literature's fan-out gains apply to weak retrievers and are pre-empted by tuned fusion; pointwise reranking of the top-20 candidates is significantly harmful on multi-gold questions (MRR −37.7pt: partially-relevant golds score below multi-channel consensus); listwise LLM reranking is statistically flat. A frozen 261-question paraphrase-robustness set shows the system is al-

ready phrasing-stable (hit@10 retention 96.5%/88.9%), explaining why query-side gains have nothing left to buy.

6. **Channel fetch depth** (O12): association-intent depth tripling gains +4.0pt cov@10 (significant) on the 300-doc slice but −2.6pt (non-significant, negative) at full-corpus scale—weak-ticket interference grows super-linearly with corpus size. Rolled back to depth 1; slice-significant ≠ corpus-significant is itself a reported finding.
7. **Generation-side context organisation**: sentence-level query-aware pruning of the retrieved context yields +6.0/+3.3pt answer-correctness directionally over two rounds (n=39/87, ns) and a significant +1.8pt faithfulness on the 87-question round; document-card headers help relevancy only; combining both mechanisms interferes (−5.1pt). Conditioning the pruner on question type is future work.
8. **Graph-side governance and bipartite diffusion**: motivated by graph-asset audits (87% of entity–relation edges are bibliographic authorship stars; the doc–entity bipartite with 1.37M links is the densest asset), three mechanisms were ablated against the deployed deep lane: (a) typed-edge governance of PPR (excluding/downweighting authored-by and affiliated-with edges, with bibliographic seed-type guards), (b) dense entity-linking fallback for seeds when the lexicon scan misses, and (c) a doc-level bipartite PPR channel. Mechanism-level checks confirm the interventions changed walk behaviour (person-seeded expansions drop 31→0 entities). Outcome: (a)+(b) are bitwise identical to baseline on every question—PPR's total contribution (∼1 question) sits below the deep lane's sensitivity floor; (c) is significantly negative (nea hit@10 −4.2pt, CI excludes zero; hard-set −2.2pt; p50 +7 s): the bipartite asset's value is already captured by the cooc channel's exact entity-pair intersection, and soft generalisation only dilutes fusion. All remain env/request-gated and disabled—the asset-layer positioning of graphs gains a sixth independent line of evidence.
9. **LLM query understanding on the deep lane**: a single LLM call per deep request classifies intent and extracts entity surfaces, overriding the template-word intent gate (natural-phrasing coverage 4/70) and feeding co-occurrence candidates. Verdict quality is high (48/48 template-assoc correctly labelled), yet on the target distribution of naturally-phrased association questions the hit count is unchanged (0 wins, 0 losses; MRR −0.04) and hard factual questions lose 2.2pt from

assoc-mechanism false triggers: the bottleneck for natural association queries is not intent detection—the association machinery itself buys nothing there. Rejected; the understanding slot stays available for trained rewriters.

10. **Deep-lane weight retune and graph-aware reranking** (both **ADOPTED**): single-variable attribution over the production tuning bundle shows the title-vector (docvec) channel actively harms the deep lane on both question families—its votes push co-occurrence winners out of the fusion pool, which reranking cannot recover (nea hit@20 −4.2pt, hard-set hit@10 −3.2pt); the doc-fusion step is the sole source of the hard-set hit@20 gain and alias expansion is strongly positive, so both stay. Complementarily, appending each candidate's top linked document entities (mention-count top-8) to the reranker input lets graph assets reach the ranking surface directly, bypassing RRF dilution—the first positive graph-side mechanism of the campaign. The deployed combination (deep lane drops docvec and the synq boost; rerank inputs carry document entities) improves every metric on both families—nea 0.896/0.917/0.583 → 0.958/1.000/0.767 (hit@10/hit@20/MRR, hit@20 saturated), hard-set 0.527/0.634/0.380 → 0.591/0.645/0.429—with 4 of 6 metrics significant under paired bootstrap and no negative metric. The normal lane keeps its own tuned configuration: per-lane weights are precisely what the dual-track design is for.
11. **Asset-layer completions and the generation-side path**: query-mined dictionary backfill adds 246 corpus-verified concepts (PWR df=88, nuclear physics df=175—core terms previously absent), taking template-association hit@10 to a saturated 1.000 (48/48) while natural-phrasing co-occurrence coverage grows 50%; alias expansion (749 validated entries) is retained as a no-cost asset. A latent budget bug (the deep lane's 120 s deadline was frozen at import time, so graph/cooc channels were silently skipped under load) is fixed and the graph channel genuinely serves deep requests again. With chains live, injecting knowledge-chain evidence into the generation context yields correctness +0.25 (CI [+0.021, +0.521], n=48)—the first significant positive result for structural graph evidence, completing the positioning: asset layer feeds retrieval, structural layer feeds generation; graphs need not enter ranking to pay their way. On a purpose-built 99-question natural-phrasing association set (template

words banned, gold = co-mention documents), the co-occurrence channel's marginal is +10.1pt hit@10 (CI [+5.1, +16.2], 10 wins 0 losses) over the identical configuration with the channel off—the asset-layer route's definitive number: template saturated at 1.000, natural phrasing +10pt at zero cost.

8.3. Engineering findings

1. **Filtered HNSW robustness**: 16% foreign-corpus vectors in a shared index cause zero steady-state degradation of the dense channel (measured over 3 days, 98 prediction files). Write-window rank jitter (±6pt arm-level on borderline questions) is a transient, not a permanent loss.
2. **LightRAG scale wall**: single-process entity merging is superlinear in graph size—82 docs/11.5k chunks/13k+ entities → ∼1 doc/hour after the first 47 docs. LLM extraction concurrency (16→70) helps; merge does not.
3. **Link-edge mass is redundant**: randomly deleting 30/60/90% of the 158,057 entity links of the test KB causes no statistically significant retrieval change (nea at 90% deletion: R@10 −2.1pt, cov@10 −2.7pt, all CI include zero; ner bitwise unchanged). Edge quality dominates edge count, and link value anchors in the structural surfaces (governance, navigation, evidence panels), not in the fusion vote.
4. **Schema binding is the root of the entity channel**: binding a corpus to the `generic` schema yields 14 canonical entities; rebinding to `nuclear_electronics` yields 1,166—an 83× difference with zero code changes, purely from the dictionary.

8.4. Limitations

1. Single-corpus domain (nuclear electronics) for the primary ablation; English benchmarks provide external validity but not domain depth.
2. Synquestion generation requires LLM access (∼40 calls per document for synq + relations); offline corpora need a pre-processing pass.
3. The intent gate is regex-based; LLM-based query classification could improve coverage but adds latency (SelRoute: 83% accuracy with regex).

Data availability

Question sets (nea/ner, 48+39 questions), evaluation scripts (paired-bootstrap scoring, scoped runners, the channel-ablation runner), and the lexicon-generation tooling are released at `https://github.com/simpleqt/knowledgenexus-artifact` (public mirror; the institutional GitLab `https://devgit.cdut.edu.cn/chenzhiqian/knowledgenexus` carries the full development history, access on request). The journal corpus itself is third-party copyrighted material; we release document-level metadata (venue/year/identifier) and chunk-level gold annotations for the evaluation slices instead. The production incident telemetry in Section 8.1 is sanitised log excerpts.

Conflict of interest

The authors declare no competing interests.

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